Build Back Better resilient infrastructure

"I am working on a project , 'Build Back Better'. This is basically about planning resilient infrastructure post disaster. Suggest me some research papers that talk about this. could you provide the exact direction for me on how do I come with better service or solution addressing real world problem."
Generated by LeapSpace

Best papers to start with

These are the most useful papers for a **Build Back Better** project on **resilient post-disaster infrastructure**, especially if you want both theory and practical guidance for designing a real-world solution [4] [20] [21].

Paper	What it gives you
Re-conceptualising "Building Back Better" to improve post-disaster recovery	The clearest overall BBB framework: it groups BBB into **risk reduction, community recovery, implementation, and monitoring/evaluation**, which is useful if you want to design a service or system rather than just describe the topic [20].
Build Back Better principles for post-disaster structural improvements	Strong for **engineering and construction choices**: building codes, inspections, cost/time, land-swap or buy-back schemes, temporary housing, and support for communities [2].
Build back better: Implementation in Victorian bushfire reconstruction	Shows BBB in practice through **hazard-based zoning, structural design improvements, permit fast-tracking, stakeholder consultation, and economic recovery support** [1].
A systematic review of factors affecting post-disaster reconstruction projects resilience	Useful for turning BBB into a **project design framework**: governance, innovation, reconstruction approach, resource management, and stakeholder expectations [21].
Supporting post-disaster social recovery to build back better	Important if you want a service that includes **community support, transparency, participation, vulnerable groups, and psychological recovery** [9].
Build Back Better: A framework for sustainable recovery assessment	Good for measuring whether your solution is actually working, using **element, structure, and function** dimensions tied to livelihood, governance, preparedness, and learning [5].
Strengthening resilience in reconstruction after extreme events – Insights from flood affected communities in Germany	Strong for the **planning, land management, infrastructure, and funding** side of reconstruction [11].

Paper	What it gives you
Shifting power through participation in post-disaster recovery	Best if your solution needs a **participation and equity** lens; it shows participation can change power relations, not just collect feedback [22].

Exact direction for a better service or solution

If you want to build a **real-world BBB service**, the evidence points to a solution that is **not just a rebuilding plan**, but a **governed, participatory, and measurable recovery system** [20] [21] [22].

1) Design the service around 4 core functions

Use the BBB framework as your backbone:

- **Risk reduction**: safer land-use, hazard zoning, stronger structural standards [1] [2] [20]
- **Community recovery**: housing, livelihoods, social cohesion, psychosocial support [9] [14] [20]
- **Implementation**: coordination, permits, funding access, contractor selection, and institutional mechanism [8] [20] [23]
- **Monitoring/evaluation**: indicators that track whether recovery is truly improving resilience over time [5] [20]

2) Build the solution as a governance system, not only a technical tool

Multiple studies show BBB succeeds when governance is strong, coordinated, and transparent [8] [11] [21]. That means your service should include:

- clear recovery authority and decision rights [8]
- structured stakeholder engagement, not one-off consultation [1] [22] [24]
- mechanisms to manage conflict, resource allocation, and accountability [25] [26] [27]

3) Make community participation operational, not symbolic

The literature warns that participation often becomes tokenistic unless power and voice are built into the process [22] [28]. A stronger solution would:

- include local residents in defining what “better” means [29] [30]
- use co-design or community-led planning [31]
- protect vulnerable groups and ensure transparency [9] [22]

4) Measure success with a practical recovery dashboard

A service is stronger if it tracks whether resilience is actually improving. The evidence supports indicators around:

- infrastructure safety and functionality [5] [11]
- governance and coordination [5] [21]
- housing and livelihood recovery [7] [14]
- social recovery and community cohesion [9] [29]

What your solution should look like in practice

A strong BBB solution for post-disaster infrastructure would combine:

1. **risk-informed planning** [1] [32]
2. **community-centered implementation** [9] [22]
3. **institutional coordination and funding pathways** [8] [11]
4. **post-recovery evaluation and learning** [5] [20]

Bottom line: the best "better service" is a **recovery platform or framework** that links **safe reconstruction, stakeholder governance, community participation, and measurable long-term resilience**, rather than a one-time rebuilding plan [20] [21] [22].

Confidence level: **High** — the retrieved papers consistently converge on the same practical direction: BBB works best when it combines governance, participation, infrastructure standards, and monitoring rather than focusing on physical rebuilding alone [9] [11] [20] [21].

References

1. Mannakkara, S., Wilkinson, S., Potangaroa, R. (2014) Build back better: Implementation in Victorian bushfire reconstruction. *Disasters*. https://doi.org/10.1111/disa.12041
2. Mannakkara, S., Wilkinson, S. (2013) Build Back Better principles for post-disaster structural improvements. *Structural Survey*. https://doi.org/10.1108/SS-12-2012-0044
4. Fernandez, G., Ahmed, I. (2019) "Build back better" approach to disaster recovery: Research trends since 2006. *Progress in Disaster Science*. https://doi.org/10.1016/j.pdisas.2019.100003
5. Zhou, B., Zhang, H., Evans, R. (2022) Build back better: A framework for sustainable recovery assessment. *International Journal of Disaster Risk Reduction*. https://doi.org/10.1016/j.ijdrr.2022.102998

7. Neeraj, S., Mannakkara, S., Wilkinson, S. (2021) Evaluating socio-economic recovery as part of building back better in Kaikoura, New Zealand. *International Journal of Disaster Risk Reduction*. https://doi.org/10.1016/j.ijdrr.2020.101930

8. Mannakkara, S., Wilkinson, S. (2016) Selecting an institutional mechanism for Building Back Better: Lessons from Victorian bushfires recovery. *International Journal of Disaster Risk Reduction*. https://doi.org/10.1016/j.ijdrr.2016.08.025

9. Mannakkara, S., Wilkinson, S. (2015) Supporting post-disaster social recovery to build back better. *International Journal of Disaster Resilience in the Built Environment*. https://doi.org/10.1108/IJDRBE-06-2013-0019

11. Birkmann, J., Schüttrumpf, H., Handmer, J., (...), Kirschbauer, L. (2023) Strengthening resilience in reconstruction after extreme events – Insights from flood affected communities in Germany. *International Journal of Disaster Risk Reduction*. https://doi.org/10.1016/j.ijdrr.2023.103965

14. Vahanvati, M., Rafliana, I. (2019) Reliability of Build Back Better at enhancing resilience of communities. *International Journal of Disaster Resilience in the Built Environment*. https://doi.org/10.1108/IJDRBE-05-2019-0025

20. Mannakkara, S., Wilkinson, S. (2014) Re-conceptualising "Building Back Better" to improve post-disaster recovery. *International Journal of Managing Projects in Business*. https://doi.org/10.1108/IJMPB-10-2013-0054

21. Charles, S.H., Chang-Richards, A.Y., Yiu, T.W. (2022) A systematic review of factors affecting post-disaster reconstruction projects resilience. *International Journal of Disaster Resilience in the Built Environment*. https://doi.org/10.1108/IJDRBE-10-2020-0109

22. Tuhkanen, H. (2023) Shifting power through participation in post-disaster recovery: A scoping review. *International Journal of Disaster Risk Reduction*. https://doi.org/10.1016/j.ijdrr.2023.104041

23. Demirci, F.S., Isik, Z. (2024) Developing a community responsive resilient contractor selection model for post-disaster reconstruction projects: a build back better approach. *Engineering, Construction and Architectural Management*. https://doi.org/10.1108/ECAM-06-2024-0802

24. Perry, E.S., Smith, S.N., Mulvaney, K.K. (2020) Designing solutions for clean water on Cape Cod: Engaging communities to improve decision making. *Ocean and Coastal Management*. https://doi.org/10.1016/j.ocecoaman.2019.104998

25. https://www.sciencedirect.com/science/article/pii/S2662998424005191

26. Chikuni, A., Kumambala, P.G., Chasukwa, M., Chinsinga, B. (2026) Are steering committees the governance fix in low- and middle-income countries (LMICs) infrastructure projects?. *Scientific African*. https://doi.org/10.1016/j.sciaf.2025.e03152

27. Okunola, O.H. (2024) Stakeholder engagement in disaster recovery: Insights into roles and power dynamics from the Ahr Valley, Germany. *International Journal of Disaster Risk Reduction*. https://doi.org/10.1016/j.ijdrr.2024.104960

28. Moore, K., Goodstadt, V. (2026) The Challenges and Opportunities of Community Participation in Disaster Recovery. *Built Environment*. https://doi.org/10.2148/benv.52.1.125

29. Livne, S., Kanyema, V., Nhlema, B., (...), Rosenthal, A. (2026) The social infrastructure of disaster recovery: policy implications for 'Build Back Better' after Cyclone Idai in Malawi. *Climate and Development*. https://doi.org/10.1080/17565529.2026.2727580

30. Collodi, J., Pelling, M., Fraser, A., (...), Di Vicenz, S. (2021) How do you build back better so no one is left behind? Lessons from Sint Maarten, Dutch Caribbean, following Hurricane Irma. *Disasters*. https://doi.org/10.1111/disa.12423

31. Paschen, J.-A., D'Arcy, Z., Evans, G., (...), Rogers, B. (2026) Community-led co-design in disaster recovery: Enabling conditions for practice. *International Journal of Disaster Risk Reduction*. https://doi.org/10.1016/j.ijdrr.2026.106373

32. Bassett, M., Wilkinson, S., Mannakkara, S. (2017) Legislation for building back better of horizontal infrastructure. *Disaster Prevention and Management*. https://doi.org/10.1108/DPM-03-2016-0054